Signed Shock Propagation and Resilience in a Global Production Network: An OECD ICIO Study

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***Abstract*—A supplier outage can damage dependent buyers and create opportunities for alternative suppliers. The earlier SIS-style model represents spreading disruption but has no explicit substitution channel. This report adds a bounded signed-state model to distinguish adverse pressure from potential opportunities, using the OECD Inter-Country Input-Output (ICIO) 2025 edition for reference year 2022. The stored baseline contains 4,050 country–industry nodes and 10.17 million positive off-diagonal links. We define targeted disruptions of the highest-flow activity, the lowest positive-flow activity, and Russia's oil-and-gas extraction node. The new signed model is reproducible, but its full-ICIO scenario outputs must be generated from the raw CSV, which is not bundled in the repository; no new scenario numbers are fabricated. Earlier SIS, threshold-cascade, and protection simulations remain as course-level comparisons. All dynamic measures are counterfactual model indices, not forecasts of actual monetary losses or gains.**

***Index Terms*—production networks, input-output analysis, signed shocks, substitution, network dynamics, SIS, resilience**

I. Introduction

A production network links suppliers to the industries that buy their intermediate inputs. When one supplier stops producing, some buyers cannot easily replace those inputs, while competing suppliers may receive redirected orders. A useful stress test should represent both sides of that event.

The original project analysed the structure of the OECD ICIO production network and used epidemic-style propagation to test how disruption could travel through supplier–buyer links. Its SIS model could describe operational and disrupted nodes, but it did not explicitly represent positive substitution opportunities. This report therefore adds a signed state with separate negative and positive components, while preserving the earlier SIS and threshold-cascade experiments as comparisons.

The report answers three practical questions: (1) what does the network structure tell us about concentration and dependence; (2) how can a small mathematical model distinguish damage from substitution opportunities; and (3) how do targeted shocks to the most and least flow-intensive industries compare in a reproducible counterfactual experiment? This is an undergraduate stress-test framework, not a calibrated economic forecasting model.

II. Data and Network Construction

A. ICIO table and node definition

The input is the regular OECD ICIO 2025-edition table for reference year 2022 [1]. It covers 80 economies plus a Rest-of-the-World (ROW) aggregate and 50 activities, giving $81 \times 50 = 4{,}050$ country–industry nodes. Monetary entries are in millions of current U.S. dollars. The raw CSV is not committed to the repository; the README records the expected file and baseline checksum.

TABLE I
Structural Snapshot of the 2022 ICIO Baseline

Measure	Baseline value
Country–industry nodes	4,050
Positive off-diagonal flows	10,166,696
Possible off-diagonal links	16,398,450
Raw directed density	62.00%
All positive intermediate flows	US$103.410 trillion
Off-diagonal flows	US$81.128 trillion
Largest weak component	3,928 nodes
Out-strength Gini	0.863
In-strength Gini	0.852

Let Z_{ij} be the intermediate flow supplied by country–industry i and used by buyer j. A directed edge $i \to j$ means supplier to purchaser. Its weight is Z_{ij}. Exact self-links are excluded from cross-node network propagation; they remain included in buyers' total input denominator where appropriate. The gross-output vector X_i is used only to weight summary indices.

B. Stored structural baseline

The following measures are from the checked-in baseline outputs, not newly recomputed from a raw data file in this revision.

Out-strength $s_i^{out} = \sum_j Z_{ij}$ measures intermediate sales supplied; in-strength $s_j^{in} = \sum_i Z_{ij}$ measures intermediate inputs purchased. Neither is GDP, gross value added, or necessarily exports. The measured graph is dense if every positive flow counts as an edge, but density falls to 5.47% at a US$1 million cutoff, 1.56% at US$10 million, and 0.345% at US$100 million. This sensitivity means that weighted flows are more informative than binary degree alone.

The highest out-strength detailed node in the stored baseline is `USA_M`, U.S. professional, scientific, and technical activities (about US$2.196 trillion); the next two are `USA_G` wholesale/retail trade (US$1.971 trillion) and `CHN_G` trade (US$1.803 trillion). The highest in-strength node is `CHN_F`, Chinese construction (about US$3.263 trillion of intermediate

inputs purchased). These are measures of flow scale, not direct estimates of shock impact.

III. Models of Disruption

A. Existing SIS and threshold-cascade baselines

In the SIS analogy, each aggregate node is either operational or disrupted, $I_j(t) \in \{0, 1\}$. For weighted influence W_{ij}, a susceptible buyer becomes disrupted with probability

$$p_j(t) = 1 - \exp\left[-\beta \sum_i W_{ij} I_i(t)\right], \tag{1}$$

while a disrupted node recovers with probability γ. The prior analysis aggregated the 50 detailed activities into 13 broad sector groups while preserving total flows and output, yielding 1,053 country–sector nodes. In its chosen high-out-strength-source experiment with $\beta/\beta_c = 1.4$, weighted SIS produced a mean peak output-weighted disrupted-node share of 10.96% (SD 8.24%), compared with 5.15% (SD 4.93%) for unweighted SIS. Cumulative disrupted-share time was 1.894 versus 0.425 share-steps. These results are stochastic, model-conditional results, not actual production losses [3], [4].

The separate threshold-cascade model assumes a buyer fails when a fixed share θ of its inputs comes from already failed suppliers. In the prior high-out-strength-seed experiment, thresholds $\theta = 0.02$, 0.05, and 0.10 caused 1,052 of 1,053 nodes to fail; at $\theta = 0.15$, 78 nodes failed; at $\theta \geq 0.30$, only the seed failed. This dramatic change illustrates sensitivity to an assumed threshold, not an empirical economic tipping point [5].

B. Why a signed model is needed

SIS is valuable for recovery and repeated outages, but a binary state cannot distinguish a buyer being harmed from an alternative supplier gaining a substitution opportunity. A signed continuous state is a minimal extension that captures these directions while keeping the mathematics accessible.

C. Input-dependency matrix

Define the cross-node flow matrix Z^{cross} by setting its diagonal to zero, and let $q_j = \sum_k Z^{all}_{kj}$ denote all intermediate input purchases of buyer j, including self-use. Then

$$D_{ij} = \begin{cases} Z^{cross}_{ij}/q_j, & q_j > 0, \\ 0, & q_j = 0. \end{cases} \tag{2}$$

Thus D_{ij} is the share of buyer j's intermediate inputs supplied by i. Its column sums are at most one, since the denominator retains internal flows and the numerator omits cross-node self-links.

D. Substitution-opportunity matrix

Let C_{ki} allocate opportunity from disrupted supplier i to potential alternative supplier k. Alternatives are restricted to the same detailed ICIO activity code in different country nodes. Weights are based on cosine similarity between supplier-to-buyer flow rows; if none of the alternatives share buyer patterns, outgoing flow strength is used as a fallback. Each source column is normalized to sum to one when alternatives exist.

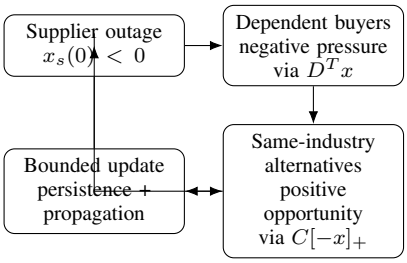


Fig. 1. Conceptual signed-shock mechanism. Negative pressure follows buyer input dependence; potential gains are assigned to plausible same-industry substitutes. Feedback is mathematical state propagation, not proof of realized market switching.

This is a transparent proxy for plausible substitution, not a measurement of unused factory capacity. ICIO's annual monetary flows do not report inventories, spare capacity, price response, or the speed at which buyers switch suppliers.

E. Signed-state dynamics

Let $x_i(t) \in [-1, 1]$. A negative value denotes adverse pressure, a positive value denotes a potential opportunity signal, and zero denotes no modeled deviation. A shock at source s initializes $x_s(0) = -h$ with all other nodes at zero. The state evolves as

$$\boxed{x(t+1) = \mathrm{clip}_{[-1,1]}\big(\lambda x(t) + \beta D^{\mathsf{T}} x(t) + \sigma C[-x(t)]_+\big)} \tag{3}$$

where $[y]_+ = \max(y, 0)$ is applied elementwise. We use teaching defaults $\lambda = 0.55$, $\beta = 0.35$, $\sigma = 0.25$, 12 steps, and full and half initial severity ($h = 1$ and $h = 0.5$).

The three terms have plain-language meanings:

- $\lambda x(t)$: pressure persists but decays over time;
- $\beta D^{\mathsf{T}} x(t)$: supplier pressure is transmitted to buyers according to input dependence, preserving its sign;
- $\sigma C[-x(t)]_+$: disrupted suppliers create positive opportunity signals for plausible substitutes.

Clipping bounds the state. The dependency-only linear part has Jacobian $J = \lambda I + \beta D^{\mathsf{T}}$. Since the row sums of D^{T} are at most one, $\lambda + \beta = 0.90 < 1$ is a sufficient contraction bound for that part under these assumptions. The substitution term is piecewise nonlinear. This model is a bounded stress test rather than a calibrated market equilibrium.

We report separate output-weighted indices using nonnegative gross output X_i:

$$N(t) = \frac{\sum_i X_i [-x_i(t)]_+}{\sum_i X_i}, \quad P(t) = \frac{\sum_i X_i [x_i(t)]_+}{\sum_i X_i}. \tag{4}$$

N measures weighted adverse pressure and P measures weighted opportunity. Neither is actual output lost or gained, GDP loss, exports, welfare, or the probability of an event. Model steps have no calendar-time interpretation.

TABLE II
TARGETED SHOCK SOURCES FROM THE STORED BASELINE

Case	Source	Meaning	Node out-strength
High-flow	USA_G	U.S. wholesale/retail trade; activity G is the largest global activity by out-strength	US$1.971T
Low-flow	CHN_C301	Chinese ship/boat building; lowest positive global activity code C301	US$22.9B
Oil case	RUS_B06	Russian crude petroleum and natural gas extraction	US$165.5B

IV. TARGETED DISRUPTION EXPERIMENTS

Three primary targets are selected using the stored baseline ranking. "Most important" and "least important" mean highest and lowest positive global outgoing intermediate-flow strength of a detailed activity code, not national importance or social value. Household own-use activity T is excluded because its recorded positive supplier flow is zero. Activity G totals about US$9.291 trillion of outgoing intermediate flows across all economies; activity C301 totals about US$83.3 billion. The signed script additionally reports a lowest-active-node sensitivity case. It runs each primary source at full and half severity and writes scenario summaries, time series, per-node effects, rankings, metadata, and figures.

The raw ICIO CSV was not present in the connected repository during this revision. Therefore, the target definitions and model code are committed, but full-dataset signed-simulation outcomes have not been computed here. The report intentionally gives no invented positive/negative scenario percentages. Generate them locally with the instructions in the README before interpreting scenario comparisons as numerical results.

A. A worked example with about 100–200 nodes

For a readable teaching view, the script exports a subnetwork containing 12 selected economies and 13 broad sector groups, up to 156 country–sector nodes. The output files are `worked_example_nodes.csv` and `worked_example_edges.csv`. The dynamic simulation still uses all 4,050 detailed nodes; the smaller export is for explanation and presentation.

For a Russia oil-and-gas scenario, the initial state of RUS_B06 is negative. Each buyer's negative pressure is computed from the share of its intermediate inputs supplied by Russia (matrix D). Alternative nodes in the same detailed activity receive a positive opportunity signal through C. Their positive state then propagates to their own buyers through $D^T x$ in subsequent steps. Which countries actually expand supply depends on capacity, policy, and prices that ICIO alone cannot determine. Also, B06 combines crude petroleum and natural gas extraction, and ROW cannot be interpreted as Venezuela specifically without country-level supporting data.

V. WHAT THE EARLIER STRESS TESTS SHOW

The prior 100-replicate SIS runs at $\beta/\beta_c = 1.4$ show a larger modelled disrupted share for dependency-weighted transmission than for the unweighted baseline (10.96% versus 5.15% peak, on average). This says that monetary dependency weights change the simulated dynamics; it does not prove that weighted SIS is more accurate.

The equal-budget intervention experiment protected 105 nodes (about 10% of eligible active nodes), halving their outgoing transmission weights. At a fixed source and the selected transmission setting, the mean peak disrupted share was 11.64% with no protection, 10.66% with random protection, 7.49% with high-PageRank protection, and 7.37% with high-out-strength protection. Mean cumulative disrupted-share time was 2.016, 1.737, 0.881, and 0.835 share-steps, respectively. These are comparisons under specified simulation assumptions, not estimates of policy impact. The signed model is a separate extension and does not inherit these numerical results.

VI. COURSE CONNECTIONS

- **Network dynamics and linear approximation:** understand edge direction, D, the signed state, and Jacobian J.
- **SIS/SIR spread models:** SIS captures operational/disrupted states with recovery; the signed-state model is continuous and distinguishes adverse pressure from opportunity.
- **Spectral thresholds and bifurcation:** future parameter sweeps can track spectral radius as β changes. This report does not claim to detect a saddle-node, transcritical, or pitchfork bifurcation.
- **Synchronization:** a follow-up could measure cross-node variance or trajectory correlation to ask whether one shock causes sectors to move together. No synchronization result is claimed here.

VII. LIMITATIONS AND CONCLUSION

The ICIO table contains annual nominal flows, not time-stamped outages. It cannot identify disruption speeds, spare capacity, inventories, elasticities, switching costs, price changes, or recovery probabilities. The structural baseline is measured data; the dynamic model is a counterfactual with assumed parameters. Output-weighted state indices must not be called GDP loss or realized output gains. The substitute matrix provides a consistent heuristic, not an empirically estimated replacement mechanism.

The revision addresses the original conceptual gap by representing both damage and potential substitution opportunity in one bounded model, and by defining high-flow, low-flow, and oil-and-gas disruption targets. It also links the analysis to network dynamics, linear approximation, SIS/SIR, spectral stability, bifurcation, and synchronization. The most defensible conclusion is that production-network topology and flow dependence help organise stress tests; realistic forecasts require

outage data, capacity constraints, and calibrated substitution responses.

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